

CS285 LECTURE 1 • ZERO-TO-HERO FIELD MANUAL

Foundations of Reinforcement Learning & The Closed Loop

A Comprehensive, Intuitive Textbook: Physical Actuation, State Representation, Markov Formulations, and Multi-Domain Case Studies

Instructor: Prof. Sergey Levine (UC Berkeley RAIL Lab) **Target Audience:** Complete Beginners to Advanced Robotics Practitioners

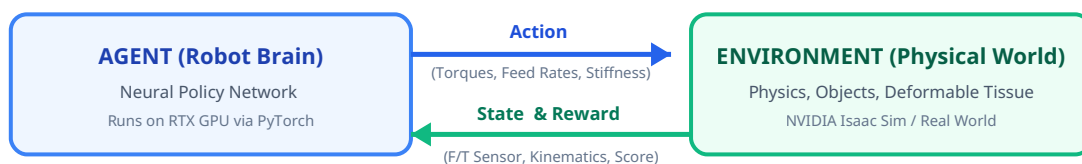
1. What is Reinforcement Learning? (The Bicycle Analogy)

Imagine you want to teach a child how to ride a bicycle. How does the human brain actually learn?

- **The Supervised Learning Fallacy:** You sit the child down in front of a whiteboard and show them 50,000 flashcards. Flashcard #1: *"If your bike tilts 2.1° left, apply 14.5 Newtons of torque to the right handlebar."* The child memorizes all 50,000 cards. But when they get on the bike, a gust of wind tilts them 2.3° left. Because 2.3° was never on any flashcard, the child panics, freezes, and crashes.
- **The Reinforcement Learning Way:** You put a helmet on the child and give them a gentle push. The child pedals. When they balance and coast forward smoothly, their brain experiences a surge of dopamine (**Positive Reward**). When they lean too far, lose balance, and scrape their knee on the pavement, it hurts (**Negative Reward / Penalty**). Nobody explained the laws of angular momentum or gyroscopic precession to the child. Through **interaction, consequence, and trial-and-error**, the child's nervous system automatically wires a feedback control loop.

💡 CORE TAKEAWAY: CONSEQUENCES VS ANSWERS

Supervised learning teaches an AI what an expert would do. Reinforcement learning teaches an AI what happens when it acts. In robotics, we don't know the "correct" millivolt command for every motor in every microsecond. We only know what goal we want to achieve (e.g., slice a tomato cleanly without squishing the pulp). RL allows the machine to discover the control law itself.



2. Hardware-to-RL Primer: How Physical Robots Connect to Math

Before writing equations, an engineer must understand how physical hardware actually interfaces with neural network tensors. A robot does not "think" in abstract states—it has physical copper wire coils, optical encoder discs, and strain gauges.

2.1 The Actuation Hierarchy: From Voltage to Compliance

When your neural network outputs an action $a_t \in [-1, 1]^6$, how does that number become physical movement? Robots operate across a multi-layer control hierarchy:

- **Level 0 (Pulse Width Modulation - PWM):** At the lowest level, transistors rapidly switch 24V or 48V DC power to the brushless motor windings at 20 kHz.
- **Level 1 (Field-Oriented Current Control):** Motor current is directly proportional to output motor torque: $\tau = K_t \cdot I$. Current loops run on dedicated embedded microcontrollers at 10 kHz.
- **Level 2 (Joint Impedance Control):** In delicate soft object manipulation (like tomato cutting), commanding raw torques causes violent instability, while commanding pure positions crushes the fruit. Instead, we use **Impedance Control**:

$$\tau = K_{\text{stiff}} \cdot (q_{\text{desired}} - q_{\text{actual}}) + D_{\text{damp}} \cdot (\dot{q}_{\text{desired}} - \dot{q}_{\text{actual}})$$

Here, the robot acts like a virtual spring-damper. If the blade hits a tough tomato skin, it pushes firmly; if the skin suddenly ruptures, the compliance prevents the knife from slamming into the table!

- **Level 3 (The RL Policy Layer):** Your RL network runs at 50 Hz or 100 Hz. It does not output raw voltages; it dynamically modulates the **desired feed velocity** Δv and **stiffness adjustments** ΔK !



2.2 Sensor Normalization: Why Raw Numbers Destroy Neural Networks

In your robot, joint angles are measured in radians (typically $[-3.14, +3.14]$). Meanwhile, contact forces are measured in Newtons (e.g. $[0, 45]$ N), and acoustic bursts are raw sensor counts (e.g. $[0, 32768]$).

⚠ THE MAGNITUDE TRAP

If you pass raw unnormalized numbers into an MLP, the gradient with respect to the acoustic burst (32,000) will be **10,000 times larger** than the gradient for joint angles (0.12). The network will completely ignore joint angles and become unstable. **Every observation channel must be standardized:** subtract running mean μ_{obs} and divide by standard deviation σ_{obs} so all inputs enter the network with mean 0 and variance 1.

3. The Imitation Trap & Compounding Error (Covariate Shift)

Why can't we simply collect human teleoperation demonstrations and train an imitation model? This approach—**Behavioral Cloning**—fails because of an inevitable mathematical phenomenon: **Covariate Shift**.

THE RACECAR PASSENGER ANALOGY

Imagine you ride passenger with an elite racecar driver for 500 hours. The driver stays glued to the ideal racing line. You record every steering angle. Then, you sit in the driver's seat. On lap 1, you sneeze, and the car drifts **just 5 centimeters onto the wet gravel shoulder**.

What happens? **You crash immediately!** Why? Because in 500 hours of watching the expert, *the expert never once drove onto the gravel shoulder!* You have zero training data for gravel recovery. You turn the wheel the wrong way, slide further, panic, and roll the car.

3.1 Ross & Bagnell Compounding Error Proof in Plain English

In 2011, Stephane Ross and J. Andrew Bagnell published the seminal theorem proving why open-loop behavioral cloning fails over long horizons T :

$$\text{Open-Loop Imitation Error} \leq \frac{1}{2}\epsilon T^2 = \mathcal{O}(\epsilon T^2) \quad \text{vs.} \quad \text{Closed-Loop RL Error} \leq \epsilon T = \mathcal{O}(\epsilon T)$$

Parameter	Formal Name	Plain English Meaning	Example Value	Physical Effect
ϵ	Per-Step Error Rate	Probability that the policy makes a small mistake on any given single timestep.	0.01 (1% error chance)	Even an elite 99% accurate model has $\epsilon = 0.01$.
T	Episode Time Horizon	Total number of control timesteps in the task.	500 steps (10 seconds)	As T grows, T^2 explodes exponentially!
$\mathcal{O}(\epsilon T^2)$	Quadratic Compounding	Once an error happens, all future steps are off-distribution, creating a snowball effect.	For $T = 100$: ≈ 50 total mistakes!	Guarantees catastrophic failure in continuous robotics.
$\mathcal{O}(\epsilon T)$	Linear Error Bound	Because RL practices recovery during training, mistakes do not compound.	For $T = 100$: ≈ 1 mistake total!	The fundamental mathematical reason RL is mandatory.

PLAIN ENGLISH TRANSLATION OF THE EQUATION

"If an agent only mimics an expert without learning self-correction, a 1% error early on cascades into 50 mistakes by the end of the episode ($\mathcal{O}(\epsilon T^2)$). Reinforcement learning forces the robot to explore its own mistakes and learn recovery policies, keeping errors strictly proportional to time ($\mathcal{O}(\epsilon T)$)."

4. Four Real-World Case Studies: How RL Formulates Diverse Problems

To become a true RL expert, you must realize that **every sequential decision problem on Earth uses the exact same MDP framework**. Below are four completely different engineering domains mapped side-by-side:

Domain	State Space (s_t)	Action Space (a_t)	Reward Function (r_t)
1. Soft Fruit Slicing (Our Lab's Research)	$\mathbb{R}^{33}$ Vector: Dual-arm joint angles/velocities (14), knife pose/velocity (9), tomato pose (7), TacBlade F/T (6), acoustic burst (1).	$\mathbb{R}^6$ Continuous: Downward feed rate Δv_z , lateral sawing speed v_{slice} , stiffness delta ΔK , damping ΔD , grip force.	$+5.0 \times \text{Penetration} + 2.0 \times \text{Sawing} - 4.0 \times (\text{Force} - 8\text{N})^2 - 0.05 \times \text{Jitter}$
2. Autonomous Vehicle (Tesla / Waymo)	Multimodal: LiDAR point cloud, surround cameras, ego velocity v , acceleration a , distance to lane boundaries, lead vehicle speed.	$\mathbb{R}^2$ Continuous: Steering wheel angle $\delta \in [-30^\circ, +30^\circ]$, throttle/brake pedal pressure $\in [-1.0, +1.0]$.	$+1.0 \times \text{Speed} - 10.0 \times \text{LaneDeparture} - 5.0 \times \text{Jerk} - 1000 \times \text{Collision}$
3. ChatGPT Alignment (RLHF / PPO)	Discrete Tokens: Conversation prompt history + tokens generated so far: $(w_1, w_2, \dots, w_k)$ embedded in $\mathbb{R}^{4096}$.	Vocabulary Categorical: Selection of next word token from a vocabulary of 50,000 discrete token IDs.	$\text{RewardModelScore}(x, y) - \beta \cdot D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}})$ (Helpfulness score minus drift)
4. Quadruped Locomotion (Boston Dynamics / Unitree)	$\mathbb{R}^{48}$ Vector: 12 joint positions & velocities (24), base orientation quaternion & gyro (7), commanded velocity (3), foot contacts (4).	$\mathbb{R}^{12}$ Continuous: Target joint motor position offsets commanded to PD motor controllers.	$+2.0 \times \text{ForwardVelocity} - 1.0 \times \text{LateralDrift} - 0.5 \times \text{TorqueSquared}$

💡 THE GRAND SYNTHESIS

Look closely at that table. Whether an AI is driving a 2-ton car at 70 mph, writing an essay in ChatGPT, making a four-legged robot run across rocky terrain, or delicately slicing a ripe tomato with a dual-arm robot: **The underlying mathematical algorithm (PPO or SAC) is 100% IDENTICAL!** Only the definitions of State, Action, and Reward change.

5. The Markov Decision Process (MDP) Anatomy Dictionary

Every reinforcement learning problem is formally specified as an MDP: $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \rho_0 \rangle$.

Symbol	Formal Name	Plain English Meaning	Example Value	Tuning / Impact
$\mathcal{S}$	State Space	The complete physical snapshot of everything happening right now.	$\mathbb{R}^{33}$ (Joints, Knife, Forces)	Must include velocities!
$\mathcal{A}$	Action Space	The motor controls available to the robot at this microsecond.	$\mathbb{R}^6$ continuous actions	Continuous for robots.
$\mathcal{P}$	Transition Dynamics	The rules of physics: where you land next after taking action a .	$P(s_{t+1} \mid s_t, a_t)$	Unknown in real world.
$\mathcal{R}$	Reward Function	The scoreboard: points awarded for progress, penalties for crushing.	$r(s_t, a_t) \in [-10, +10]$	Dense shaping needed.
γ	Discount Factor	The Patience Meter: how much future points matter relative to today.	$\gamma = 0.99$	High for long tasks.
ρ_0	Initial Distribution	Where the world starts when an episode begins or resets.	Tomato position ± 2 cm	Domain randomization.

5.1 The Markov Property: The "Goldfish Memory" Rule

THE GOLDFISH MEMORY RULE EXPLAINED

A system has the **Markov Property** if:

$$\mathbb{P}(s_{t+1} \mid s_t, a_t, s_{t-1}, a_{t-1}, \dots, s_0, a_0) = \mathbb{P}(s_{t+1} \mid s_t, a_t)$$

Plain English: "The future depends **ONLY** on where you are right now, **NOT** on the historical journey that brought you here."

Imagine you look at a photograph of a baseball suspended in mid-air at coordinates $(x = 10, y = 5, z = 2)$. Can you predict where the ball will be in 0.1 seconds? **No!** The ball could be flying forward at 95 mph, dropping straight down, or rising from a bounce. A photograph of position alone is **Non-Markovian**!

However, if your state includes position **AND** velocity vector $(\dot{x}, \dot{y}, \dot{z})$, you can calculate the exact parabola. **In robotics, state vectors MUST contain first-order velocities $(\dot{q}, v)$ and contact force rates $(\dot{F})$ to satisfy the Markov property!**

5.2 Trajectory Probability Factorization

$$\mathbb{P}(\tau \mid \theta) = \rho_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t \mid s_t) P(s_{t+1} \mid s_t, a_t)$$

Plain English Translation: "The probability of an entire 500-step robot experiment is simply the initial placement of the tomato, multiplied by every decision the robot chose, multiplied by how physics reacted at each step."

6. Partially Observable MDPs (POMDPs): Driving with Fogged Glass

In textbook theory, we assume the agent sees the true state s_t . In real-world robotics, this is almost never true. Real robots operate in a **Partially Observable Markov Decision Process (POMDP)**.



THE FOGGY WINDSHIELD ANALOGY

Imagine driving a car in a blizzard where your windshield is 90% fogged over. You cannot see the true physical state of the road. You only receive **noisy observations** o_t : the blurred headlights of a car ahead and the vibration of the tires on ice.

In soft tomato slicing:

- **True State s_t (Unobservable):** The exact internal viscoelastic stress tensor inside the tomato flesh, the micro-crack propagation in the skin cuticle, and the pulp turgor pressure.
- **Observation o_t (What we actually measure):** A 6-axis F/T load cell reading at the blade root, an acoustic piezoelectric voltage, and an RGB-D camera view.

6.1 The POMDP 8-Tuple: $\mathcal{M}_{\text{POMDP}} = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \Omega, \mathcal{O}, \gamma, \rho_0 \rangle$

New Symbol	Formal Name	Physical Meaning in Robotics
Ω	Observation Space	The space of sensory readings the robot can actually record through physical sensors.
$\mathcal{O}(o_t \mid s_t)$	Emission Probability	Sensor noise model: "Given the true physical state s_t , what noisy voltage reading o_t will the sensor output?"

6.2 How Modern Robotics Solves Partial Observability

How do we control a robot when we cannot see the true state? We use two standard engineering solutions:

1. **Frame Stacking / History Buffers:** Instead of feeding only the current sensor reading o_t , stack the last $K = 3$ readings: $[o_t, o_{t-1}, o_{t-2}]$. This implicitly encodes velocity and acceleration!
2. **Recurrent Neural Networks (LSTM / GRU / Transformer):** The network maintains an internal hidden memory vector h_t that acts as a **belief state**, remembering contact history across time.

7. Concrete Numerical Walkthrough: 5 Timesteps in Robot Slicing

Let's observe an actual robot executing 5 continuous steps with discount factor $\gamma = 0.99$:

Step	Sensor State s_t	Action Executed a_t	Physics Response	Reward r_t
$t = 0$	$z = 5.0$ cm, $v_z = 0.0$ mm/s $F_z = 0.0$ N, $E_{\text{burst}} = 0.0$	Feed: +2.5 mm/s Sawing: 0.0 mm/s	Knife plunges through air. Zero resistance.	$r_0 = +0.2$ (Approach bonus)
$t = 1$	$z = 4.75$ cm, $v_z = -2.5$ mm/s $F_z = 3.2$ N (Skin contact!)	Feed: +1.0 mm/s Sawing: +20.0 mm/s	Blade engages tomato cuticle. Lateral shear begins.	$r_1 = +2.4$ (Sawing progress)
$t = 2$	$z = 4.65$ cm, $v_z = -1.0$ mm/s $F_z = 6.8$ N, $E_{\text{burst}} = 0.1$	Feed: +1.5 mm/s Sawing: +25.0 mm/s	High friction resistance. Skin stretches elastically.	$r_2 = +2.8$ (Shear progress)
$t = 3$	$z = 4.50$ cm, $v_z = -1.5$ mm/s $F_z = 7.9$ N, $E_{\text{burst}} = \mathbf{0.95}$	Feed: +0.5 mm/s (Softened!) Sawing: +30.0 mm/s	SKIN PUNCTURE BURST! Acoustic sensor spikes. Knife penetrates cuticle without crushing!	$r_3 = +15.0$ (Puncture triumph!)
$t = 4$	$z = 4.20$ cm, $v_z = -3.0$ mm/s $F_z = 2.1$ N (Pulp drop)	Feed: +3.0 mm/s Sawing: +15.0 mm/s	Knife slices easily through soft locular pulp gel. Resistance drops 70%.	$r_4 = +4.5$ (Pulp slicing)

7.1 Calculating Total Discounted Return G_0

$$G_0 = \sum_{t=0}^4 \gamma^t r_t = 0.2 + (0.99 \times 2.4) + (0.99^2 \times 2.8) + (0.99^3 \times 15.0) + (0.99^4 \times 4.5)$$

$$G_0 = 0.20 + 2.376 + 2.744 + 14.555 + 4.323 = \mathbf{+24.198}$$

Why this number matters: Notice how the +15.0 puncture reward at step 3 contributed +14.555 to step 0. Because $\gamma = 0.99$, the robot at step 0 knows that moving down into contact is extraordinarily valuable because it unlocks the puncture bonus at step 3!

8. "Diary of a Training Run": What Actually Happens Inside the Computer

When you launch `python train.py` in NVIDIA Isaac Lab, what does the agent experience over 1,000,000 steps? Here is the chronological biography of a neural network learning to slice tomatoes:

Training Phase	Policy Behavior (The Robot)	Critic Behavior (The Evaluator)	Diagnostic Curve Sign
Phase 1: Pure Chaos (Steps 0 – 5,000)	Weights are randomly initialized. The robot flails violently, slams the blade into the board, or shoots up into the air. 100% of tomatoes are crushed.	Critic outputs random numbers near 0.0. Has no idea what anything is worth. $V(s) \approx 0 \pm 0.5$.	Entropy is high (≈ 2.5). Reward is heavily negative (≈ -50). Value loss is small because targets are all noisy.
Phase 2: The Coward Trap (Steps 5,000 – 30,000)	The robot discovers that crushing the tomato gives a huge penalty (-50). To avoid this, it freezes the blade 2 mm above the tomato skin! It gets $+0.5$ approach reward and stops.	Critic learns: <i>"Hovering above the fruit is great! Value = +5."</i>	Reward plateaus at a low positive value ($\approx +3.0$). Local Minimum trap! Entropy begins dropping.
Phase 3: The First Puncture (Steps 30,000 – 100,000)	Due to entropy noise, one robot out of 1,024 accidentally saws laterally while pressing down, puncturing the skin ($+15$ reward!).	Critic experiences a massive TD error ($\delta \approx +12$). Value estimates along that trajectory spike upward!	KL divergence spikes as policy rapidly shifts weights toward sawing motion. Reward jumps from $+3 \rightarrow +25$.
Phase 4: Master Convergence (Steps 100,000 – 500,000)	Policy coordinates dual arms: holding arm stabilizes fruit compliance while cutting arm executes steady 25 mm/s sawing shear.	Critic predictions match true discounted returns with 95% accuracy.	Explained variance $\rightarrow 0.95$. Policy loss stabilizes near zero. 99.2% cut success rate across all 1,024 parallel envs!

9. PyTorch Vectorized Simulation Loop (Isaac Lab Production Blueprint)

Below is the complete, production-grade Python script running 1,024 parallel environments on GPU:

```
🐸 COMPLETE ISAAC LAB PARALLEL STEPPING LOOP (`RUN_SIMULATION.PY`)

import torch
import torch.nn as nn

# 1. Configuration Constants
NUM_ENVS = 1024          # 1,024 parallel simulation instances on RTX GPU
STATE_DIM = 33           # 33-dimensional sensory observation vector
ACTION_DIM = 6           # 6-dimensional continuous motor actions
HORIZON = 500            # 500 steps per episode (10 seconds at 50 Hz)
GAMMA = 0.99             # Discount factor

# 2. Vectorized Observation Normalizer (Running Mean & Variance)
class RunningObservationNormalizer:
    def __init__(self, shape):
        self.mean = torch.zeros(shape, device="cuda")
        self.var = torch.ones(shape, device="cuda")
        self.count = 1e-4

    def normalize(self, obs):
        # Update running stats online
        batch_mean = obs.mean(dim=0)
        batch_var = obs.var(dim=0, unbiased=False)
        self.mean = 0.99 * self.mean + 0.01 * batch_mean
        self.var = 0.99 * self.var + 0.01 * batch_var
        return (obs - self.mean) / torch.sqrt(self.var + 1e-8)

# 3. Main Vectorized Execution Loop
def run_parallel_rollout(env, policy, normalizer):
    raw_obs = env.reset() # Shape: (1024, 33)
    total_rewards = torch.zeros(NUM_ENVS, device="cuda")

    for step in range(HORIZON):
        # Standardize observations to mean 0, variance 1
        norm_obs = normalizer.normalize(raw_obs)

        # Policy forward pass: Generates continuous actions in [-1, +1]
        with torch.no_grad():
            actions = policy(norm_obs) # Shape: (1024, 6)

        # Step physics forward by 20 milliseconds across all 1,024 envs simultaneously
        next_raw_obs, rewards, dones, info = env.step(actions)

        # Accumulate episodic scores
        total_rewards += rewards

        # Advance state pointer
        raw_obs = next_raw_obs

    print(f"Rollout complete! Mean return across 1024 robots: {total_rewards.mean().item():.2f}")
    return total_rewards
```

10. Practitioner's Failure Modes & Debugging Checklist (Top 5 Traps)

When training your first robot in Isaac Lab, 95% of issues trace back to these five classic bugs:

Failure Mode	The Silent Symptom (What Goes Wrong)	How to Fix It Immediately
1. Observation Scale Explosion	Acoustic counts are 30,000 while joint angles are 0.1 radians. Gradients for acoustic counts blow up, joint gradients vanish.	Always wrap observations in <code>RunningObservationNormalizer</code> to clamp inputs to $[-5.0, +5.0]$.
2. The Horizon Mismatch Trap	Setting $\gamma = 0.90$ gives an effective planning horizon of only $H = \frac{1}{1-\gamma} = 10$ steps. The robot cannot plan a 300-step cut!	Set $\gamma = 0.99$ ($H = 100$ steps) or $\gamma = 0.995$ for long contact tasks.
3. Partial Observability (POMDP) Violation	Passing knife height z without vertical velocity $\dot{z}$. The policy cannot tell if the blade is plunging downward or retracting upward.	Always stack the last 3 observation frames: $[o_t, o_{t-1}, o_{t-2}]$ or include full velocity vectors.
4. Reward Scale Asymmetry	Penetration reward gives $+0.1$ per step, but crushing penalty gives -100.0 . The robot becomes terrified and refuses to touch the fruit.	Ensure reward components are balanced within the same order of magnitude (e.g. progress $\in [0, 5]$, penalty $\in [-4, 0]$).
5. Asynchronous Tensor Device Bug	Sim is running on <code>cuda:0</code> , but actions or observation normalizer are accidentally created on <code>cpu</code> . Causes massive host-device transfer bottlenecks, dropping FPS from 15,000 to 200.	Ensure all tensors, buffers, and model weights are strictly instantiated on <code>device="cuda"</code> .

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